Team: T123 **Github:** `https://github.com/Anoop-Parashar/COL7361-Assignments`
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Preliminaries

In this section, we delineate certain preliminaries that include special notation and definitions. We leverage this notation scheme in later sections.

Let $G = (V, E, p)$ be a directed graph where $V = \{1, 2, \ldots, n\}$ represents forest regions and $E \subseteq V \times V$ represents potential fire-spread routes between regions. Each edge $(u, v) \in E$ is associated with a probability function $p : E \to (0, 1]$ such that $p(u, v)$ denotes the probability that fire spreads from region u to region v.

Let $A_t \subseteq V$ denote the set of regions that are burning at time step t. The initial set of burning regions is denoted by A_0, and the fire spreads in discrete time steps according to the following stochastic rule: if $u \in A_t$ and $(u, v) \in E$, then region v catches fire at time $t + 1$ with probability $p(u, v)$ independently of other edges.

Let A_∞ denote the set of regions that eventually burn once the propagation of fire terminates.

To mitigate the spread of fire, we may block a subset of routes $R \subseteq E$, where $|R| \leq k$ for a given blocking budget k. Removing these routes yields a modified graph

$$G_R = (V, E \setminus R, p).$$

Let

$$\sigma(R) = \mathbb{E}[|A_\infty|]$$

denote the expected number of burned regions when the fire spreads on graph G_R starting from the initial set A_0. Equivalently,

$$\sigma(R) = \sum_{u \in V} \Pr[u \in A_\infty \mid A_0, G_R].$$

where $Pr[.]$ is the probability operator.

Additionally, we note the direct similarity of this problem with the Influence Minimization via Edge Blocking problem under the Independent Cascade (IC) model, as introduced by Xie et al. [2025].

Task 1: Optimization Formulation

Our goal is to determine the set of routes to block such that the expected number of burned regions is minimized under the budget constraint. Mathematically, this can be written in the form of the following optimization problem:

$$R^* = \arg\min_{R \subseteq E} \quad \sigma(R) \tag{1}$$

$$\text{s.t.} \quad |R| \leq k. \tag{2}$$

Here, the optimization function is $\sigma(R)$, the decision variable is R, *i.e*, the set of blocked routes, and the constraint is $|R| \leq k$.

Task 2: NP-Hardness

We show that our objective of minimizing fire spread across forest regions under a blocking budget, k, is NP-hard via a reduction from the well-known **Maximum Coverage Problem (MCP)** [Hochbaum and Pathria, 1998].

Proof. In the Maximum Coverage problem, we are given a universe of items $U = \{e_1, \ldots, e_n\}$, a collection of subsets $S_1, \ldots, S_m \subseteq U$, and an integer k. The objective is to choose at most k subsets whose union covers the maximum number of elements. This problem is known to be NP-hard.

Using the notation and definitions given in Preliminaries, we can, without loss of generality, assume the vertex set of G to be defined as $V = \{s\} \cup \{v_1, \ldots, v_m\} \cup \{u_1, \ldots, u_n\}$, where s is a source node, each v_i represents set S_i (we call these "set-nodes"), and each u_j represents element e_j (we call these "element-nodes"). We connect s to each set-node using an edge, *i.e*, $\forall i \in [1, m], (s, v_i) \in E$. Additionally, we connect each set-node with each constituent element-node, *i.e.*, $e_j \in S_i \implies (v_i, u_j) \in E$. Since set-nodes can correspond to arbitrarily chosen sets of elements e_j, we do not lose any generality by defining G in this manner. Additionally, we set the propagation probability for each edge in E to 1 in order to model the worst-case scenario for this stochastic process, *i.e.*, that given enough time, all nodes in G get burned. We assume A_0 to contain only the source node s.

Since the fire spreads independently from s to all v_i, if we block an edge (s, v_i), we would prevent the spread of fire to node v_i, and subsequently, to all of the element-nodes reachable from it. Therefore, under a budget constraint of k, we can say that minimizing the spread of fire is equivalent to removing k set-nodes from G such that the number of corresponding element-nodes that are protected is maximized. In other words, we must choose k set-nodes which cover the maximum number of element-nodes, which is exactly the maximum coverage problem (MCP). Since MCP is NP-hard, the route blocking problem, which we have proven to be equivalent to MCP, is also NP-hard.

□

Task 3: Structural Properties

Monotonicity

We prove that $\sigma(R)$ exhibits non-increasing monotonicity, *i.e.*, for two sets of blocked routes R_1 and R_2, $R_1 \subseteq R_2 \implies \sigma(R_2) \leq \sigma(R_1)$.

Proof. Let R_1 and R_2 be two sets of blocked routes such that $R_1 \subseteq R_2 \subseteq E$. Clearly, $\sigma(R_1)$ and $\sigma(R_2)$ represent the expected fire spread in both cases, respectively. As defined in Preliminaries, $G_{R_1} = (V, E \setminus R_1, p)$ and $G_{R_2} = (V, E \setminus R_2, p)$. Since $R_1 \subseteq R_2$, that means $E \setminus R_2 \subseteq E \setminus R_1$. Therefore, G_{R_2} is a subgraph of G_{R_1}.

This means that each edge path in G_{R_2} is contained in G_{R_1} but not vice versa. In other words, every directed edge path in G_{R_2} that makes node $v \in V$ reachable from A_0 must also exist in G_{R_1}. Thus, the set of reachable nodes from A_0 in G_{R_2} must be a subset of reachable nodes in G_{R_1}. Mathematically,

$$|A_\infty(G_{R_2})| \leq |A_\infty(G_{R_1})| \tag{3}$$

Using linearity of expectation,

$$\mathbb{E}\Big[|A_\infty(G_{R_2})|\Big] \leq \mathbb{E}\Big[|A_\infty(G_{R_1})|\Big] \tag{4}$$

$$\implies \sigma(R_2) \leq \sigma(R_1) \tag{5}$$

Hence, as the number of blocked routes increases, the expected fire spread never increases. □

Submodularity

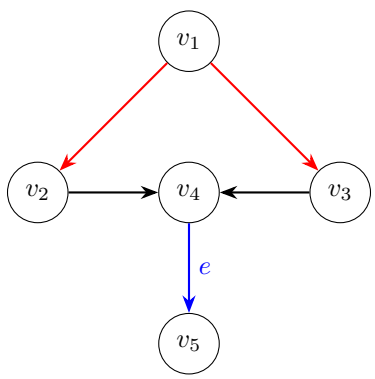

Figure 1: Graph used to provide a counter-example to disprove submodularity. Red edges belong to R_2, while the blue edge is $e = (v_4, v_5)$.

We prove that $\sigma(R)$ is NOT submodular with the help of a counter-example.

Proof. Firstly, in order for $\sigma(R)$ to be submodular, it must satisy the following inequality:

$$\sigma(R_1 \cup \{e\}) - \sigma(R_1) \geq \sigma(R_2 \cup \{e\}) - \sigma(R_2) \tag{6}$$

where R_1 and R_2 are two sets of blocked routes such that $R_1 \subseteq R_2 \subseteq E$ and $e \notin R_2$.

Now, we consider a directed graph with $V = \{v_1, v_2, \ldots, v_5\}$ and $E = \{(v_1, v_2), (v_1, v_3), (v_2, v_4), (v_3, v_4), (v_4, v_5)\}$, as shown in Figure 1. For simplicity, $\forall e \in E, p(e) = 1$ and $A_0 = \{v_1\}$. Assume that $R_1 = \varnothing$ and $R_2 = \{(v_1, v_2), (v_1, v_3)\}$. Clearly, $R_1 \subseteq R_2$. Additionally, $\sigma(R_1) = 5$ and $\sigma(R_2) = 1$. Let us consider edge $e = (v_4, v_5) \notin R_2$. We can see that $\sigma(R_1 \cup \{e\}) = 4$ since removing edge e will completely disconnect node v_5 from the source node. However, for R_2, the blocked routes already block all out-edges from the source node,

meaning that the fire cannot spread to any non-source nodes to begin with. As a result, adding e to this set will not affect the spread of fire in the graph. In other words, $\sigma(R_2 \cup \{e\} = 1$. Clearly,

$$\sigma(R_1 \cup \{e\}) - \sigma(R_1) = 4 - 5 = -1 \tag{7}$$

$$\sigma(R_2 \cup \{e\}) - \sigma(R_2) = 1 - 1 = 0 \tag{8}$$

Since $-1 < 0$,

$$\sigma(R_1 \cup \{e\}) - \sigma(R_1) < \sigma(R_2 \cup \{e\}) - \sigma(R_2) \tag{9}$$

which violates the inequality condition given in Inequality 6. Hence, the above counterexample proves that $\sigma(R)$ is not submodular.

□

Task 4: Algorithm Design

We leverage the GreedyReplace algorithm proposed by Xie et al. [2025], adapted for edge blocking (IMIN-EB). The algorithm operates in three phases and uses DESCE (Dominator-based Estimation of Spread Change for Edges) to estimate the decrease in expected spread caused by blocking each candidate edge in a single pass per sample.

Algorithm 1 DESCE(G, A_0, θ) — Expected Influence Decrease Estimation

Require: Graph $G = (V, E, p)$, seed set A_0, number of samples θ
Ensure: $\Delta[u, v]$: estimated decrease in expected spread if edge (u, v) is blocked

```
Initialize Δ[(u, v)] ← 0 for all (u, v) ∈ E
for t = 1 to θ do
    Construct empty virtual graph G^E
    Add virtual root v*
    for each seed s ∈ A_0 do
        Add edge v* → s in G^E
    end for
    for each edge (u, v) ∈ E do
        if Bernoulli(p(u, v)) = 1 then
            Create virtual node w_uv
            Add edges u → w_uv and w_uv → v
        end if
    end for
    T ← dominator tree of G^E rooted at v*
    for each node x in T do
        sub[x] ← number of original nodes in the subtree of x
    end for
    for each sampled virtual node w_uv do
        Δ[(u, v)] ← Δ[(u, v)] + sub[w_uv]/θ
    end for
end for
return Δ
```

Algorithm 2 GreedyReplace-EB(G, A_0, k, θ, V_h) — Edge Blocking

Require: Graph $G = (V, E, p)$, initially burning nodes A_0, budget k, samples θ, h-hop filter V_h
Ensure: Set of blocked routes $R \subseteq E$, $|R| \leq k$

```
// Phase 1: Greedy over seed out-edges
C_R ← {(s, v) : s ∈ A_0, (s, v) ∈ E, v ∈ V_h}
R ← ∅,   order ← []
for i = 1 to min(|C_R|, k) do
    Δ ← DESCE(G \ R, A_0, θ, V_h)
    (u*, v*) ← arg max_{(u,v)∈C_R\R} Δ[(u, v)]
    R ← R ∪ {(u*, v*)};   order.append((u*, v*))
end for
// Phase 2: Replace in reverse insertion order
for e ∈ reverse(order) do
    R ← R \ {e}                                          ▷ Temporarily remove e
    Δ ← DESCE(G \ R, A_0, θ, V_h)
    (u*, v*) ← arg max_{(u,v)∉R} Δ[(u, v)]
    if (u*, v*) = e then
        R ← R ∪ {e};   break
    else
        R ← R ∪ {(u*, v*)}
    end if
end for
// Phase 3: Fill remaining budget
while |R| < k do
    Δ ← DESCE(G \ R, A_0, θ, V_h)
    (u*, v*) ← arg max_{(u,v)∉R} Δ[(u, v)]
    if no candidate found then break
    end if
    R ← R ∪ {(u*, v*)}
end while
return R
```

Computational Complexity

Let $n = |V|$, $m = |E|$, k be the budget, θ be the number of DESCE samples, and $\alpha(m, n)$ denote the inverse Ackermann function from union-find.

DESCE. Each of the θ iterations samples all m edges in $O(m)$, builds the virtual graph G^E with at most $2m$ additional nodes and $2m$ additional edges, then computes the dominator tree in $O(m \cdot \alpha(m, n))$ using the Lengauer–Tarjan algorithm [Lengauer and Tarjan, 1979]. The bottom-up subtree count requires one pass over the BFS order in $O(m)$. Hence one call to DESCE costs $O(\theta \cdot m \cdot \alpha(m, n))$.

Phase 1. Let $d_s^{\text{out}} = |C_R|$ denote the total number of seed out-edges. Phase 1 calls DESCE once per greedy round for $\min(d_s^{\text{out}}, k)$ rounds, giving total cost:

$$O\big(\min(d_s^{\text{out}},\, k) \cdot \theta \cdot m \cdot \alpha(m, n)\big)\,.$$

Phase 2. Phase 2 iterates over at most $\min(d_s^{\text{out}}, k)$ edges in reverse order, calling DESCE once per edge over *all* candidates. In the worst case this doubles the Phase 1 cost, so Phase 2 is also $O\big(\min(d_s^{\text{out}},\, k) \cdot \theta \cdot m \cdot \alpha(m, n)\big)$.

Phase 3. Phase 3 runs at most $k - \min(d_s^{\text{out}}, k)$ additional rounds, each calling DESCE once, contributing at most $O(k \cdot \theta \cdot m \cdot \alpha(m, n))$.

Overall. Combining all phases, the total complexity of GreedyReplace-EB is:

$$\boxed{O(k \cdot \theta \cdot m \cdot \alpha(m, n))}.$$

Approximation Guarantees

There is no guarantee under IC wherein the estimated expected decrease in fire spread $\sigma(R)$ is within, say, $(1 - 1/e)$ of the optimal solution, as is the case under the LT model. This is due to the fact that under the IC model, $\sigma(R)$ is *not* submodular (proven in Task 3). This means that it is possible that blocking two edges together leads to a much larger decrease in expected fire spread than blocking either edge alone, which directly violates the diminishing returns property associated with submodularity. This can be seen with the help of the counter-example provided in Task 3. We can, however, prove that the DESCE estimator is an unbiased estimator of the expected decrease in spread and that the estimation error associated with it decreases exponentially with the number of Monte Carlo simulated samples.

Proof. From Algorithm 1, the number of nodes that would become unreachable from A_0 if the edge $e = (u, v)$ is blocked in a randomly sampled graph G^E is represented by $X_e(G^E) = |\text{sub}[w_{uv}]|$. The expected decrease in spread can then be represented as $\Delta(e) = \mathbb{E}[X_e(G^E)]$ since each G^E is sampled independently under the IC model. Let the DESCE estimator estimate this expected decrease using θ Monte Carlo-sampled graphs. This estimator can be represented as $\tilde{\Delta}(e) = \frac{1}{\theta}\sum_{i=1}^{\theta} X_e(G_i^E)$. Now,

$$\mathbb{E}[\tilde{\Delta}(e)] = \mathbb{E}\left[\frac{1}{\theta}\sum_{i=1}^{\theta} X_e(G_i^E)\right] \tag{10}$$

$$= \frac{1}{\theta}\sum_{i=1}^{\theta} \mathbb{E}[X_e(G_i^E)] \tag{11}$$

$$= \Delta(e) \tag{12}$$

Therefore, $\tilde{\Delta}(e)$ is an unbiased estimator.

Now, since all sampled graphs G_i^E are i.i.d and $0 \leq X_e(G_i^E) \leq |V|$, we can use Hoeffding's inequality [Hoeffding, 1963] as follows:

$$Pr\Big[|\tilde{\Delta}(e) - \mathbb{E}[\tilde{\Delta}(e)]| \geq \epsilon\Big] \leq 2\exp\left[\frac{-2\theta\epsilon^2}{(|V| - 0)^2}\right] \tag{13}$$

$$\implies Pr\Big[|\tilde{\Delta}(e) - \Delta(e)]| \geq \epsilon\Big] \leq 2\exp\left[\frac{-2\theta\epsilon^2}{|V|^2}\right] \tag{14}$$

where $\epsilon > 0$ is an error threshold. Clearly, the estimation error for each edge is bounded by an exponential factor which decreases as the number of Monte Carlo simulations (θ) increases. Note that this bound only tells us that the expected decrease estimation will be more accurate with higher values of θ but it does not prove anything about whether the solutions that arise out of these increasing accurate estimations will be optimal or not.

□

References

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